

ReplicationPad1d#

[https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad1d.html#tor](https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad1d.html#torch.nn.ReplicationPad1d)

Description:

Pads the input tensor using replication of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 2-tuple, uses `(padding_left\text{padding\_left}padding_left, padding_right\text{padding\_right}padding_right)` Note that the output dimensions must remain positive. Input: `(C,Win)(C, W_{in})(C,Win)` or `(N,C,Win)(N, C, W_{in})(N,C,Win)`. Output: `(C,Wout)(C, W_{out})(C,Wout)` or `(N,C,Wout)(N, C, W_{out})(N,C,Wout)`, where

Function Signature

```
class torch.nn.ReplicationPad1d(padding)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReplicationPad1d(2)
>>> input = torch.arange(8, dtype=torch.float).reshape(1, 2, 4)
>>> input
tensor([[[0., 1., 2., 3.],
         [4., 5., 6., 7.]]])
>>> m(input)
tensor([[[[0., 0., 0., 1., 2., 3., 3., 3.],
         [4., 4., 4., 5., 6., 7., 7., 7.]]]])
>>> # using different paddings for different sides
>>> m = nn.ReplicationPad1d((3, 1))
>>> m(input)
tensor([[[[0., 0., 0., 0., 1., 2., 3., 3.],
         [4., 4., 4., 4., 5., 6., 7., 7.]]]])
```

Detailed Documentation

Documentation

Pads the input tensor using replication of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 2-tuple, uses $(padding_left, padding_right)$. Note that the output dimensions must remain positive.

Input: (C, W_{in}) or (N, C, W_{in})

Output: (C, W_{out}) or (N, C, W_{out}) , where

$$W_{out} = W_{in} + padding_left + padding_right$$